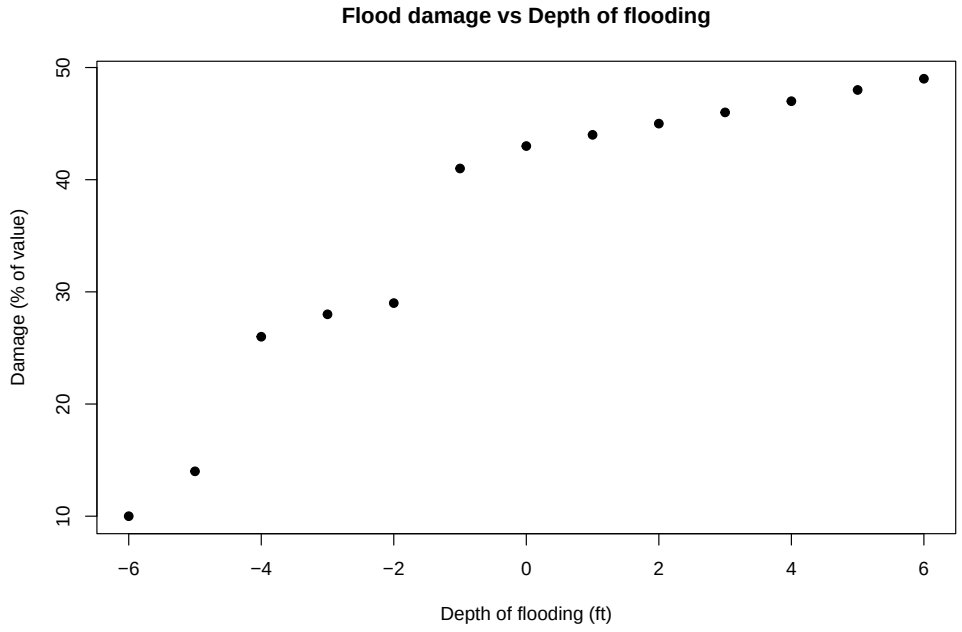
Tutorial 9 - Solutions

1. We have

$$\begin{aligned} RSS &= \sum (y_i - \hat{\alpha} - \hat{\beta} x_i)^2 \\ &= \sum \left(y_i - \bar{y} + \frac{s_{xy}}{s_{xx}} \bar{x} - \frac{s_{xy}}{s_{xx}} x_i \right)^2 \\ &= \sum \left((y_i - \bar{y}) - \frac{s_{xy}}{s_{xx}} (x_i - \bar{x}) \right)^2 \\ &= s_{yy} - 2 \frac{s_{xy}}{s_{xx}} s_{xy} + \frac{s_{xy}^2}{s_{xx}^2} s_{xx} \\ &= s_{yy} - \frac{s_{xy}^2}{s_{xx}}. \end{aligned}$$

2. (a) The scatter plot does not show a clear linear relationship between payments and claims.



(b)

$$n = 13, \sum x = 0, \sum x^2 = 182, \sum y = 470, \sum y^2 = 19118$$
$$\sum xy = 577.$$
$$s_{xx} = 182, s_{yy} = 19118 - \frac{470^2}{13} = 2125.692$$
$$s_{xy} = 577.$$

So $\hat{\beta} = \frac{577}{182} = 3.17$ and $\hat{\alpha} = \frac{470}{13} - 0 = 36.15$. Therefore the fit is

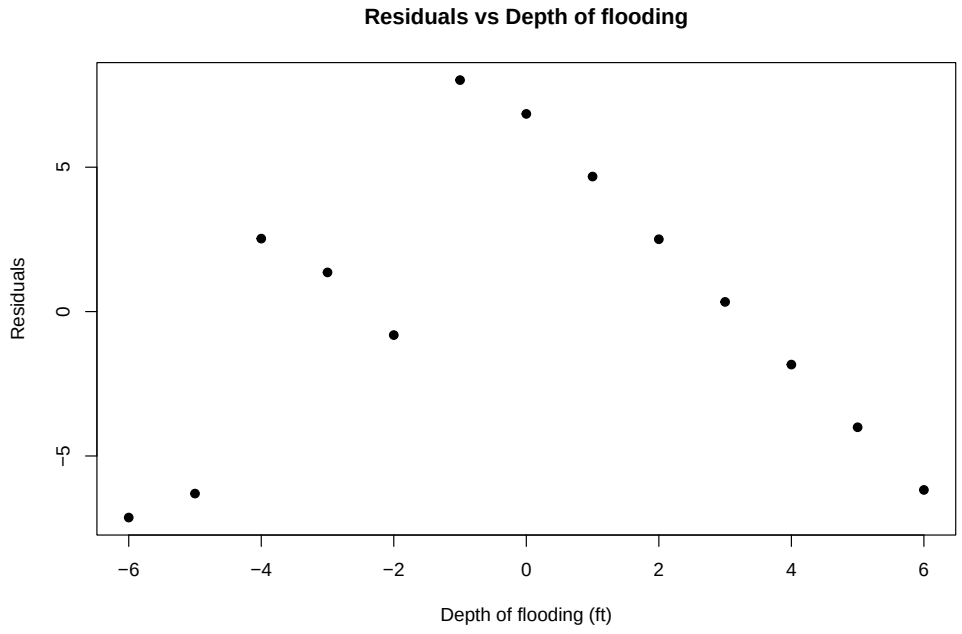
$$\hat{y} = 36.15 + 3.17x.$$

(c) Assume that the response variables (Y_i) are independent normally distributed with common variance, σ^2 say.

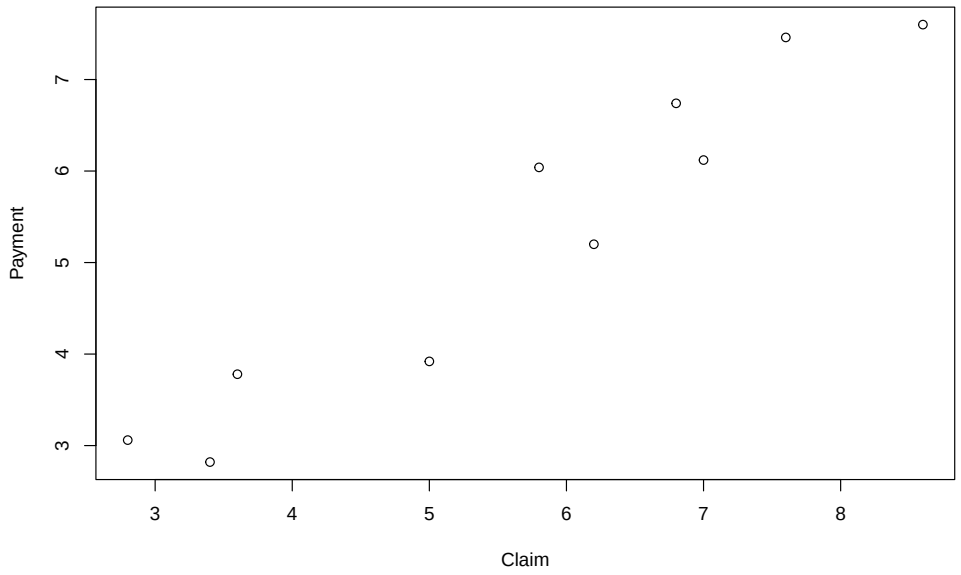
$$\hat{\sigma}^2 = \frac{1}{n-2}\left(s_{yy} - \frac{s_{xy}^2}{s_{xx}}\right) = \frac{1}{11}\left(2125.692 - \frac{577^2}{182}\right) = 26.947.$$

The estimated standard error of $\hat{\beta}$ is $\sqrt{\frac{\hat{\sigma}^2}{s_{xx}}}$ This calculates as $\sqrt{\frac{26.947}{182}} = 0.385$. The appropriate t distribution is t_{11} which has "cut-off" points for 2.5% as 2.201. Therefore the 95% confidence interval for β is $3.17 \pm (2.201 \times 0.385)$, i.e. $(2.32, 4.02)$.

(d) The residuals can be calculated in the usual way, $y_i - \hat{y}_i$, and a graph of these against x gives the following:



This is not a random pattern and so there is some doubt as to the suitability of the regression model.



3. (a) There is a strong association between payments and claims.

   (b) From question 2, we have $s_{xx} = 33.776, s_{xy} = 29.8008$.
   Therefore

$$\hat{\beta} = \frac{s_{xy}}{s_{xx}} = \frac{29.8008}{33.776} = 0.8823$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = \frac{52.74}{10} - \frac{29.8008 \times 56.8}{10} = 0.2625$$

   So the equation of the line of best fit is $y = 0.8823x + 0.2625$.

   (c) We know that

$$\frac{\hat{\beta} - \beta}{\sqrt{\frac{\hat{\sigma}^2}{s_{xx}}}} \sim t_{n-2}$$

   For 95% confidence interval, we note that $t_8(2.5\%) = 2.306$.
   So we solve

$$\frac{\hat{\beta} - \beta}{\sqrt{\frac{\hat{\sigma}^2}{s_{xx}}}} = \pm 2.306.$$

   Now

$$\hat{\sigma}^2 = \frac{1}{n-2}\left(s_{yy} - \frac{s_{xy}^2}{s_{xx}}\right) = \frac{1}{8}\left(28.63524 - \frac{29.8008^2}{33.776}\right) = 0.2927$$

so

$$\sqrt{\frac{\hat{\sigma}^2}{s_{xx}}} = \sqrt{\frac{0.2927}{33.776}} = 0.09309$$

For 95% CI, we have $\beta = 0.8823 \pm 2.306 \times 0.09309 = (0.6676, 1.0970)$.

Because 0 is not included in this interval, it is unlikely that we would accept the hypothesis that $\beta = 0$. We will do a hypothesis test anyway.

$$H_0 : \beta = 0 \quad v \quad H_1 : \beta \neq 0$$

Note that the test statistic is

$$\frac{\hat{\beta} - 0}{0.09309} = \frac{0.8823 - 0}{0.09309} = 9.478.$$

Using R, the p-value for this test statistic is $2 \times 0.0000063238 = 0.000012648$. This is overwhelming evidence against H_0 and so tests using a significance level as low as 0.0012648% would reject the null hypothesis.

(d) 95% confidence limits for the expected (mean) Y at $X = x_0$ are given by:

$$(\alpha + \beta x_0) \pm t_{n-2}(2.5\%)\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}}}$$

This gives

$$(0.2625 + 0.8823 \times 7.80) \pm 2.306 \times \sqrt{0.2927}\sqrt{\frac{1}{10} + \frac{(7.80 - 5.68)^2}{33.776}} = (6.54, 7.75).$$

95% confidence limits for the individual Y at $X = x_0$ are given by:

$$(\alpha + \beta x_0) \pm t_{n-2}(2.5\%)\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}} + 1}$$

This gives

$$(0.2625+0.8823\times 7.80)\pm 2.306\times\sqrt{0.2927}\sqrt{\frac{1}{10} + \frac{(7.80 - 5.68)^2}{33.776} + 1} = (5.76, 8.53).$$

(e)

$$R^2 = \frac{SS_{REG}}{SST} = \frac{s_{xy}^2}{s_{xx}s_{yy}} = \frac{29.8008^2}{33.776 \times 28.63524} = 0.9182.$$

Therefore, the proportion of total variance explained by the regression model is 91.82%. This is very high.

4. (a) For the data $\sum x = 34$, $\sum x^2 = 155$, $\sum y = 138.8$, $\sum y^2 = 2603.38$, $\sum xy = 634.6$. We first calculate

$$s_{xx} = \sum x_i^2 - \frac{1}{8}\left(\sum x_i\right)^2 = 10.5$$
$$s_{yy} = \sum y_i^2 - \frac{1}{8}\left(\sum y_i\right)^2 = 195.2$$
$$s_{xy} = \sum x_i y_i - \frac{1}{8}\left(\sum x_i\right)\left(\sum y_i\right) = 44.7.$$

We now obtain $\hat{\beta} = \frac{s_{xy}}{s_{xx}} = 4.257$ and $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = -0.74225$. The regression line is

$$\hat{E}(Y) = 4.257x - 0.742.$$

To obtain a 95% CI for β first calculate

$$RSS = s_{yy} - \frac{s_{xy}^2}{s_{xx}} = 4.906.$$

Since under repeated sampling $RSS \sim \sigma^2\chi_6^2$ and $\hat{\beta} \sim N(\beta, \frac{\sigma^2}{s_{xx}})$, it follows that

$$\frac{(\hat{\beta} - \beta)\sqrt{s}_{xx}}{\sqrt{\frac{RSS}{6}}} \sim t_6.$$

We now obtain the 95% CI for β:

$$\left(\hat{\beta} - t_{6,0.025}\sqrt{\frac{RSS}{6s_{xx}}}, \hat{\beta} + t_{6,0.025}\sqrt{\frac{RSS}{6s_{xx}}}\right).$$

Substituting in the values ($t_{6,0.025} = 2.447$), we obtain (3.574, 4.440) for the CI. This located well away from zero indicating strong evidence for a positive association between glucose concentration and size of culture.

(b) A plot of the data looks fairly linear. Also the coefficient of determination

$$r^2 = \frac{s_{xy}^2}{s_{xx}s_{yy}} = 0.975$$

indicates that the linear model is capturing much of the variability in the values of y.

(c) Predicted value at $x = 3.7$ is

$$\hat{y} = \hat{\alpha} + \hat{\beta}x = -0.742 + 4.257 \times 3.7 = 15.0$$

(or e^{15} for the actual count).

(d) This change is given by the slope β of the regression line, so we predict that the count will change (on the log scale) by $\hat{\beta} = 4.257$ for every unit of increase in glucose concentration (or by a number of $e^{4.257} = 70.6$ for the actual count).

Exam Style Question

(a) From the plot there appears to be a strong negative association between sleep and age.

[2 marks]

(b) Given that

$$\sum x_i = 560, \sum y_i = 6652, \sum x_i^2 = 25188, \sum y_i^2 = 2955918, \sum x_i y_i = 243444$$

calculate the fitted line for the regression of y on x.

Using the statistics given we have

$$\begin{aligned} s_{xx} &= 25188 - \frac{560^2}{15} = 4281.33, \\ s_{xy} &= 243444 - \frac{(560)(6652)}{15} = -4897.33, \\ s_{yy} &= 2955918 - \frac{6652^2}{15} = 5977.73. \end{aligned}$$

[1 mark]

We also have

$$\begin{aligned} \bar{x} &= \frac{560}{15} = 37.33 \\ \bar{y} &= \frac{6652}{15} = 443.47. \end{aligned}$$

This gives us the estimates:

$$\begin{aligned} \hat{\beta} &= \frac{s_{xy}}{s_{xx}} = \frac{-4897.33}{4281.33} = -1.144 \\ \hat{\alpha} &= \bar{y} - \hat{\beta}\bar{x} = \frac{6652}{15} - (1.144)\frac{560}{15} = 486.176 \end{aligned}$$

[1 mark]

Therefore the equation we found is:

$$\hat{y} = 486.176 - 1.144x.$$

(c) The standard error of estimate for an individual with $x = 60$ is

$$\hat{\sigma}\sqrt{\frac{1}{15} + \frac{(60 - 37.3333)^2}{4281.33} + 1}.$$

where

$$\hat{\sigma}^2 = \frac{1}{n-2}\left(s_{yy} - \frac{s_{xy}^2}{s_{xx}}\right) = \frac{1}{13}\left(5977.73 - \frac{(-4897.33)^2}{4281.33}\right) = 28.91.$$

Thus the standard error is

$$\sqrt{28.91}\sqrt{\frac{1}{15} + \frac{(60-37.3333)^2}{4281.33} + 1} = 6.465.$$

The estimate is

$$\hat{y}(60) = \hat{\alpha} + \hat{\beta}(60) = 486.176 - 1.144(60) = 417.536.$$

Using that $t_{n-2,0.025} = 2.160$ we have the observed confidence interval

$$(417.536 - 2.160 \times 6.465, 417.536 + 2.160 \times 6.465) = (403.572, 431.5).$$

[3 marks]

(d) For $x = 60$ the suggested equation gives:

$$\text{Sleeping time} = 480 - 60 = 420.$$

This is consistent with the confidence interval in (c) so we have insufficient evidence to contradict the claim.

[2 marks]

[Total 12 marks]